YEAR 7 SCIENCE · SPACE UNIT

Ready for the CAT: revision worksheet

Fill in the boxes as we go through the slides. Answers are shown on the slide after each task – tick or fix yours in a different colour.

Name: Class: Date:

1 Quick-fire eight

Starter · from memory, no booklet

1 How long does Earth take to spin once on its axis?

2 How many days for Earth to travel once around the Sun?

3 By how many degrees is Earth's axis tilted?

4 How many days from one full Moon to the next?

5 One astronomical unit (1 AU) is how many kilometres?

6 How far does light travel in one second?

7 Which layer of the Sun do we see as its "surface"?

8 Is Neptune a terrestrial planet, a gas giant or an ice giant?

Score: / 8

2 Which layer am I?

Layers · pairs

Part A · Earth Word bank: Inner core · Outer core · Mantle · Crust

A1 I'm liquid iron and nickel, and my movement makes Earth's magnetic field.

A2 I'm only 5–70 km thick and I'm broken into tectonic plates.

A3 I'm solid iron and nickel at the very centre of the planet.

A4 I'm semi-solid rock about 2,900 km thick.

Part B · The Sun Word bank: Core · Radiative zone · Convective zone · Photosphere · Chromosphere · Corona

B1 I'm the Sun's visible surface, where sunspots form.

B2 Nuclear fusion happens in me, at about 15 million °C.

B3 I'm the outermost layer, at over 1 million °C.

B4 Energy takes about 100,000 years to cross me as light.

B5 I'm the thin layer of atmosphere just above the visible surface.

B6 Hot plasma rises and sinks through me, carrying energy outward.

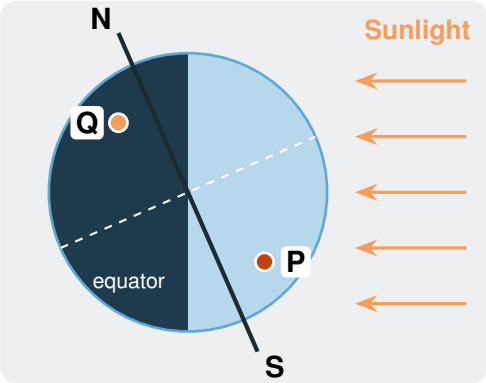
Challenge: write each set of layers in order, from the centre out.

Earth → → → →

Sun → → → → → →

3 Read the diagram: places P and Q

Day, night and seasons · on your own



1 Which place, P or Q, is in daylight right now? [1]

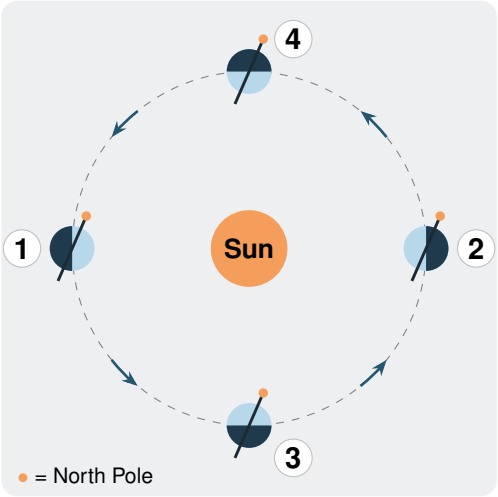
2 Which place is having summer? Use the tilt of the axis to explain. [2]

3 Roughly which month of the year could this be? [1]

4 Fix this sentence: "Earth *revolves* on its axis once every 24 hours." [1]

4 Earth's year in four positions

Day, night and seasons · pairs



a At which position is Melbourne having summer?

b Follow the arrows. At which position is Melbourne having spring?

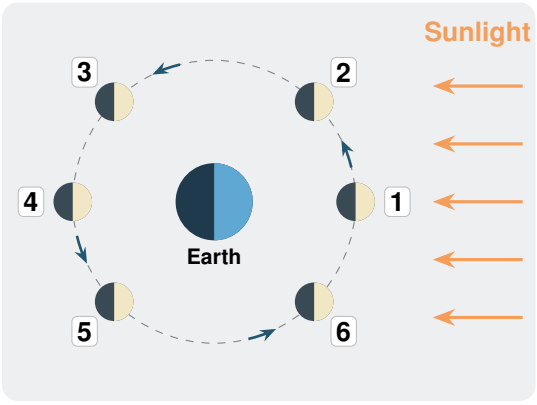
c At which two positions are day and night equal everywhere?

d Japan is in the Northern Hemisphere. At which position is Japan having winter?

e "Melbourne is cold in July because Earth is furthest from the Sun then." Earth really *is* furthest in July. So what is wrong with this reasoning?

5 The Moon at positions 1 to 6

Moon phases · on your own, then check




(i)


(ii)


(iii)


(iv)

Word bank: New · Waxing crescent · First quarter · Waxing gibbous · Full · Waning gibbous · Third quarter · Waning crescent

a Name the phase seen from Earth at:

position 2 _____

position 5 _____

b Which drawing, (i) to (iv), shows the Moon at position 3 as seen from Melbourne? _____

c Which drawing shows the Moon at position 6 as seen from Melbourne? _____

d Name the phase seen from Earth at:

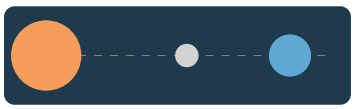
position 1 _____

position 4 _____

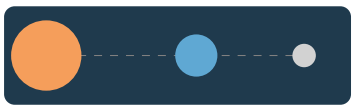
e "The dark part of a crescent Moon is Earth's shadow." Correct this statement.

6 Three line-ups: spring or neap?

Tides · pairs


(i) Sun · Moon · Earth


(ii) Moon at right angles


(iii) Sun · Earth · Moon

Orange = Sun, blue = Earth, grey = Moon (not to scale)

1 Which line-up(s) cause spring tides? Name the Moon's phase in each. [2]

2 Which causes neap tides? Why are neap tides smaller? [2]

3 There is a new Moon on 2 November. About when are the next neap tides? [1]

4 High tide at a beach is at 6 am. When is the next high tide, and which motion causes it? [2]

5 If the Moon vanished, would the oceans stop having tides? Explain. [2]

7 Your turn: five calculations

Calculations · show all working

Data: Sun diameter 1,392,000 km · Moon diameter 3,474 km · Earth to Moon 384,400 km · Mars to Sun 228,000,000 km · Mars 1.52 AU · Saturn 9.54 AU · Neptune 30.1 AU · 1 AU = 150,000,000 km · speed of light = 300,000 km/s
Every time: ① write the rule ② put the numbers in ③ calculate ④ answer with units, rounded as asked

1 How many times wider is the Sun than the Moon? Nearest whole number. [2]

Rule and working

Answer (with units)

2 Saturn is 9.54 AU from the Sun. Calculate this distance in km. [2]

Rule and working

Answer (with units)

3 How long does sunlight take to reach Mars? Give your answer in seconds, then in minutes and seconds. [2]

Rule and working

Answer (with units)

4 How many times farther from the Sun is Neptune than Mars? Nearest whole number. [2]

Rule and working

Answer (with units)

5 Challenge: mission control radios an astronaut on the Moon, who replies instantly. What is the shortest wait between sending and hearing the reply? [2]

Rule and working	Answer (with units)

8 Sort, classify, explain

Solar system · pairs

1 Put them in order from the Sun: Uranus · Mars · Saturn · Mercury · Neptune · Earth · Jupiter · Venus

1	2	3	4	5	6	7	8

My own mnemonic: _____

2 Odd one out: Mercury, Venus, Saturn, Mars. Which, and why?

3 Who am I?

a) The first planet beyond the asteroid belt.

b) An ice giant, 7th from the Sun, spinning on my side.

c) The hottest planet, though not the closest to the Sun.

4 "The asteroid belt is what's left of a planet that exploded." Give the scientific explanation instead.

5 Where is the Kuiper Belt? Name a dwarf planet found there.

9 Explain it: two marks, two points, joined by "because"

Explain it · on your own

Sentence frame: "This happens because . . . This means that . . . "

A Planet temperatures. Mercury is closer to the Sun than Venus. Yet Mercury's night side drops to -180 °C, while Venus stays at about 465 °C day and night. Explain why. [2]

B Space debris. A fleck of paint once chipped a window on the International Space Station. Explain how something so tiny could do damage, and why the crew could not steer around it. [2]

C Sky knowledge. A friend says the dark patches along the Milky Way are “just empty gaps”. Correct them, and explain how Aboriginal astronomers use these dark shapes as a calendar. [2]

10 A four-mark diagram answer

Draw and explain · on your own

Japan is in the Northern Hemisphere. Using a labelled diagram, explain why Japan has its summer in July, the same month that Melbourne has winter. Refer to your diagram in your explanation. [4]

What earns the marks: ☐ Sun in the middle, Earth drawn twice ☐ axis tilted the *same way* both times, N and S labelled ☐ which hemisphere leans toward the Sun ☐ Sun higher, more direct rays, longer days ☐ *not* distance

Diagram (use a pencil and ruler)

Explanation (“In my diagram, at position A . . . ”)

Peer mark: ____ / 4

11 Exit: rate yourself before the CAT

Exit ticket · tick one box per row

Topic	Booklet	I can. . .	Confident	Shaky	Need help
Layers of Earth and Sun	§1.2, §1.4 p.3–5	name and order all 4 + 6 layers	☐	☐	☐
Day, night, seasons	§2 p.8–10	read an orbit diagram; explain tilt	☐	☐	☐
Moon phases	§3.1 p.11	name a phase; pick the Melbourne view	☐	☐	☐
Tides	§3.2 p.12–13	spring vs neap; two tides a day	☐	☐	☐
Calculations	§1.1, §1.4, §5.1	ratios, AU to km, light time, with working	☐	☐	☐
Solar system	§5 p.20–22	order and classify; the asteroid belt	☐	☐	☐
Moon, debris, sky	§1.3, §6.3, §7	explain each in two linked sentences	☐	☐	☐

My two topics to revise tonight: 1. _____ 2. _____